

1. Write a python program to create an empty data frame.

```
import pandas as pd
df=pd.DataFrame()
print(df)
```

```
Empty DataFrame
Columns: []
Index: []
```

2. Write a python program to add marks and attendance of 5 students.

```
import pandas as pd
data={'Marks':pd.Series([45,48,65,69,72],index=['Neel','Shashank','Dishant','Shree','Shreyash']),'Attendance':pd.Series([75,69,71,100,49],index=['Neel','Shashank','Dishant','Shree','Shreyash'])}
df=pd.DataFrame(data)
print(df)
```

	Marks	Attendance
Neel	45	75
Shashank	48	69
Dishant	65	71
Shree	69	100
Shreyash	72	49

3. Write a python program to add new columns.

```
import pandas as pd
data={'Marks':pd.Series([45,48,65,69,72],index=['Neel','Shashank','Dishant','Shree','Shreyash']),'Attendance':pd.Series([75,69,71,100,49],index=['Neel','Shashank','Dishant','Shree','Shreyash'])}
df=pd.DataFrame(data)
print(df)
print("\nAdding a new column:\n")
df['Age']=pd.Series([19,18,17,17,20],index=['Neel','Shashank','Dishant','Shree','Shreyash'])
df['Date of birth']=pd.Series(['24 April','6 Jan','5 November','18 April'],index=['Neel','Shashank','Dishant','Shree','Shreyash'])
print(df)
```

	Marks	Attendance
Neel	45	75
Shashank	48	69
Dishant	65	71
Shree	69	100
Shreyash	72	49

Adding a new column:

	Marks	Attendance	Age	Date of birth
Neel	45	75	19	24 April
Shashank	48	69	18	6 Jan
Dishant	65	71	17	5 November
Shree	69	100	17	18 August
Shreyash	72	49	20	24 April

4.Retrieve the data only of student number 1.

```
print("Retrieving details of Neel:\n")
print(df.loc['Neel'])
```

Retrieving details of Neel:

Marks	45
Attendance	75
Age	19
Date of birth	24 April
Name: Neel, dtype: object	

5.Delete a column using del function.

```
print("Deleting the total column using del function:\n")
df.pop('Date of birth')
print(df)
```

Deleting the total column using del function:

	Marks	Attendance	Age
Neel	45	75	19
Shashank	48	69	18
Dishant	65	71	17
Shree	69	100	17
Shreyash	72	49	20

6.Calculate mean along the columns.

```
import pandas as pd
mydictionary={'Names': ['Neel', 'Shashank', 'Amol', 'Dishant'],
'Physics':[68, 74, 77, 78], 'Chemistry':[84, 56, 73, 69], 'Algebra':[78,
88, 82, 87]}
dfmarks=pd.DataFrame(mydictionary)
print('DataFrame:\n')
print(dfmarks)
mean = dfmarks.mean()
print('\nMean:\n')
print(mean)
```

DataFrame:

	Names	Physics	Chemistry	Algebra
0	Neel	68	84	78

1	Shashank	74	56	88
2	Amol	77	73	82
3	Dishant	78	69	87

Mean:

```
Physics      74.25
Chemistry    70.50
Algebra      83.75
dtype: float64
```

C:\Users\Student\AppData\Local\Temp\ipykernel_12904\2166369846.py:7:
FutureWarning: Dropping of nuisance columns in DataFrame reductions
(with 'numeric_only=None') is deprecated; in a future version this
will raise TypeError. Select only valid columns before calling the
reduction.

```
mean = dfmarks.mean()
```

7.Adding a row.

```
import pandas as pd
df1= pd.DataFrame(
[['Neel', 68, 84, 78, 96],
['Shashank', 74, 56, 88, 85],
['Dishant', 77, 73, 82, 87]],
columns=['Name','Physics','Chemistry','Algebra','Calculus'])
df2=pd.DataFrame(
[['Shree', 72, 67, 91, 83],
['Shreyash', 78, 69, 87, 92]],
columns=['Name','Physics','Chemistry','Algebra','Calculus'])
frames=[df1, df2]
df=df1.append(df2, ignore_index=True)
print("df1:\n",df1)
print("\ndf2:\n",df2)
print("\ndf:\n",df)
```

df1:

	Name	Physics	Chemistry	Algebra	Calculus
0	Neel	68	84	78	96
1	Shashank	74	56	88	85
2	Dishant	77	73	82	87

df2:

	Name	Physics	Chemistry	Algebra	Calculus
0	Shree	72	67	91	83
1	Shreyash	78	69	87	92

df:

	Name	Physics	Chemistry	Algebra	Calculus
0	Neel	68	84	78	96
1	Shashank	74	56	88	85

2	Dishant	77	73	82	87
3	Shree	72	67	91	83
4	Shreyash	78	69	87	92

8.Create and display a one-dimensional array of an item in a shop containing an array of data.

```
import pandas as pd
ds = pd.Series([2, 4, 6, 8, 10])
print(ds)
```

```
0    2
1    4
2    6
3    8
4   10
dtype: int64
```

9.Convert a Panda module Series of products to Python list products and it's type.

```
import pandas as pd
ds = pd.Series([2, 4, 6, 8, 10])
print("Pandas Series and type")
print(ds)
print(type(ds))
print("Convert Pandas Series to Python list")
print(ds.tolist())
print(type(ds.tolist()))
```

```
Pandas Series and type
0    2
1    4
2    6
3    8
4   10
dtype: int64
<class 'pandas.core.series.Series'>
Convert Pandas Series to Python list
[2, 4, 6, 8, 10]
<class 'list'>
```

10.Add,subtract,multiple and divide two Pandas Series of items.

```
import pandas as pd
ds1 = pd.Series([2, 4, 6, 8, 10])
ds2 = pd.Series([1, 3, 5, 7, 9])
ds = ds1 + ds2
print("Add two Series:")
print(ds)
print("Subtract two Series:")
ds = ds1 - ds2
print(ds)
```

```

print("Multiply two Series:")
ds = ds1 * ds2
print(ds)
print("Divide Series1 by Series2:")
ds = ds1 / ds2
print(ds)

```

Add two Series:

```

0    3
1    7
2   11
3   15
4   19

```

dtype: int64

Subtract two Series:

```

0    1
1    1
2    1
3    1
4    1

```

dtype: int64

Multiply two Series:

```

0    2
1   12
2   30
3   56
4   90

```

dtype: int64

Divide Series1 by Series2:

```

0    2.000000
1    1.333333
2    1.200000
3    1.142857
4    1.111111

```

dtype: float64

11.Convert a dictionary of items to a Pandas series.

```

import pandas as pd
d1 = {'a': 'milk', 'b': 'biscuit', 'c': 'soap', 'd': 'paper',
      'e': 'chips'}
print("Original dictionary:")
print(d1)
new_series = pd.Series(d1)
print("Converted series:")
print(new_series)

```

Original dictionary:

```
{'a': 'milk', 'b': 'biscuit', 'c': 'soap', 'd': 'paper', 'e': 'chips'}
```

Converted series:

```
a    milk
```

```
b    biscuit
c      soap
d     paper
e     chips
dtype: object
```

12.Change the data type of given a column or a Series.

```
import pandas as pd
s1 = pd.Series(['100', '200', 'python', '300.12', '400'])
print("Original Data Series:")
print(s1)
print("Change the said data type to numeric:")
s2 = pd.to_numeric(s1, errors='coerce')
print(s2)
```

```
Original Data Series:
0      100
1      200
2    python
3    300.12
4      400
dtype: object
Change the said data type to numeric:
0      100.00
1      200.00
2         NaN
3      300.12
4      400.00
dtype: float64
```

13.Convert Series of coloured shirts list to one Series.

```
import pandas as pd
s = pd.Series([
    ['Red', 'Green', 'White'],
    ['Red', 'Black'],
    ['Yellow']])
print("Original Series of list")
print(s)
s = s.apply(pd.Series).stack().reset_index(drop=True)
print("One Series")
print(s)
```

```
Original Series of list
0    [Red, Green, White]
1         [Red, Black]
2         [Yellow]
dtype: object
One Series
0         Red
```

```
1    Green
2    White
3     Red
4    Black
5    Yellow
dtype: object
```

14.Sort a given Series.

```
import pandas as pd
s = pd.Series(['100', '200', 'python', '300.12', '400'])
print("Original Data Series:")
print(s)
new_s = pd.Series(s).sort_values()
print(new_s)
```

Original Data Series:

```
0    100
1    200
2  python
3  300.12
4    400
dtype: object
0    100
1    200
3  300.12
4    400
2  python
dtype: object
```

15.Create a dataframe from dict of ndarrays.

```
import pandas as pd
info = {'ID' :[101, 102, 103], 'Department' :
['B.Sc', 'B.Tech', 'M.Tech',]}
info = pd.DataFrame(info)
print (info)
```

```
   ID Department
0  101      B.Sc
1  102      B.Tech
2  103      M.Tech
```

16.How will you create an empty DataFrame in Pandas?

```
import pandas as pd
info = pd.DataFrame()
print (info)
```

```
Empty DataFrame
Columns: []
Index: []
```

17.How will you add a column to a pandas DataFrame?

```
import pandas as pd
info = {'one' : pd.Series([1, 2, 3, 4, 5], index=['a', 'b', 'c', 'd', 'e']),
        'two' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f'])}
info = pd.DataFrame(info)
print ("Add new column by passing series")
info['three']=pd.Series([20,40,60],index=['a','b','c'])
print (info)
print ("Add new column using existing DataFrame columns")
info['four']=info['one']+info['three']
print (info)
```

Add new column by passing series

	one	two	three
a	1.0	1	20.0
b	2.0	2	40.0
c	3.0	3	60.0
d	4.0	4	NaN
e	5.0	5	NaN
f	NaN	6	NaN

Add new column using existing DataFrame columns

	one	two	three	four
a	1.0	1	20.0	21.0
b	2.0	2	40.0	42.0
c	3.0	3	60.0	63.0
d	4.0	4	NaN	NaN
e	5.0	5	NaN	NaN
f	NaN	6	NaN	NaN

18.How to get the items of series A not present in series B?

```
import pandas as pd
p1 = pd.Series([2, 4, 6, 8, 10])
p2 = pd.Series([8, 10, 12, 14, 16])
p1[~p1.isin(p2)]
```

```
0    2
1    4
2    6
dtype: int64
```

19.How to get frequency counts of unique items of a series?

```
import pandas as pd
import numpy as np
p= pd.Series(np.take(list('pqrstu'), np.random.randint(6, size=17)))
p = pd.Series(np.take(list('pqrstu'), np.random.randint(6, size=17)))

p.value_counts()
```



```
q    5
p    4
t    3
s    2
u    2
r    1
dtype: int64
```

20.How to convert a numpy array to a dataframe of given shape?

```
import pandas as pd
import numpy as np
p = pd.Series(np.random.randint(1, 7, 35))
# Input
p = pd.Series(np.random.randint(1, 7, 35))
info = pd.DataFrame(p.values.reshape(7,5))
print(info)
```

	0	1	2	3	4
0	1	1	2	2	4
1	2	2	1	5	6
2	2	4	2	3	6
3	4	6	1	6	1
4	2	4	5	1	3
5	4	4	5	3	1
6	1	3	3	3	4